Problem Statement : Remaining Useful Life (RUL) Prediction for Lithium-Ion Batteries

Business Context

Battery warranty, lifecycle cost, and customer confidence are key differentiators in the EV and Energy Storage market. Accurate Remaining Useful Life (RUL) prediction can significantly reduce warranty risk, enable predictive maintenance, and open new service-based revenue streams.

Problem Statement

Design a data-driven model to predict the Remaining Useful Life (RUL) of a lithium-ion battery pack based on operational and environmental parameters.

Scope and Expectations

Participants are expected to:

- Develop an algorithm / model that predicts RUL in terms of cycles, months, or energy throughput
- Use given or simulated datasets including:
    - Voltage, current, temperature
    - State of Charge (SoC), Depth of Discharge (DoD)
    - Charge - discharge cycles
- Demonstrate:
    - Model logic (ML / physics-based / hybrid)
    - Accuracy trends and degradation behaviour

Expected Output

- RUL prediction model (code or logic flow)
- Visualization of degradation trends
- Dashboard or simple UI (optional)

Evaluation Criteria

- Prediction accuracy and robustness
- Technical approach and explainability
- Scalability for EV / BESS applications
- Business relevance (warranty, AMC, SaaS potential)

Business Impact

- Enables extended warranty offerings
- Reduces field failures and service costs
- Potential Battery Health Analytics product

Remaining Useful Life (RUL) Prediction for Lithium-Ion Battery Pack in Electric Light Commercial Vehicle

1. Background

With the rapid adoption of electric mobility in India, battery life and reliability have become key factors influencing the total cost of ownership (TCO) for commercial vehicles. Accurate prediction of Remaining Useful Life (RUL) of lithium-ion batteries can help enable predictive maintenance, optimize warranty policies, and improve fleet utilization.

The New Energy Division of Endurance Technologies Limited is developing battery systems for electric commercial vehicles. Predicting battery degradation under real-world Indian operating conditions is therefore critical.

This Hackathon challenge invites participants to develop a data-driven or physics-informed algorithm to estimate the Remaining Useful Life (RUL) of a lithium-ion battery pack used in an electric commercial vehicle application.

2. System Configuration

Participants should consider the following battery pack and vehicle configuration as the base case.

Battery Chemistry

- **Cell Chemistry** : Lithium Iron Phosphate (LFP)
- **Nominal Cell Voltage** : ~3.2 V
- **Typical Cycle Life** : ~3000–4000 cycles (at 80% DoD under standard conditions)

Battery Pack Configuration

- **Pack Energy Capacity** : 35 kWh
- **Nominal Pack Voltage** : ~350 V
- **Battery Architecture** : Modular pack design
- **Cooling System** : Active Liquid Cooling
- **Operating SOC Window** : 10% – 90%

Battery Management System (BMS)

The BMS measures and provides the following parameters:

- Cell voltage
- Pack voltage
- Pack current
- Cell temperature
- State of Charge (SoC)
- Charge–discharge cycles

- Depth of Discharge (DoD)
- Charging rate (C-rate)

Participants may assume that historical operational data is available.

3. Vehicle Application

The battery pack is used in an electric light commercial vehicle (LCV)

Vehicle Category

Electric Light Commercial Vehicle (eLCV)

Typical applications include:

- Last-mile delivery
- Urban logistics
- Fleet operations

Vehicle Parameters

Parameter	Value
Motor Power Rating	60 kW peak
Continuous Motor Power	30 kW
Vehicle Gross Weight	~2.5 tons
Average Daily Distance	120 - 150 km
Charging Strategy	Overnight AC charging

4. Driving Cycle Assumption

Participants should use the Modified Indian Driving Cycle (MIDC – Annexure A) or similar urban driving patterns representative of Indian commercial vehicle operation.

Typical characteristics:

Parameter	Value
Average Speed	25 - 30 km/h
Stop-and-Go Driving	High
Frequent Acceleration	Yes
Regenerative Braking	Enabled
Ambient Temperature Range	20°C - 45°C

Participants should incorporate Indian climatic and operational conditions, which influence battery degradation.

5. Problem Statement

Develop an algorithm or model capable of predicting the Remaining Useful Life (RUL) of the battery pack based on operational data and degradation trends.

The model should estimate:

- Remaining cycle life
- Remaining calendar life
- Remaining energy capacity before reaching End-of-Life (EOL)

End-of-Life Definition

Battery is considered End-of-Life when usable capacity falls to 80% of initial capacity.

6. Scope and Expectations

Participants may use one or a combination of the following approaches:

- Machine learning / data-driven modelling
- Physics-based degradation modelling
- Hybrid models
- Statistical trend analysis

The model should consider degradation drivers such as:

- Depth of discharge
- Charge/discharge rate
- Temperature exposure
- Number of cycles
- Driving patterns

Participants should demonstrate how their model predicts capacity fade over time.

7. Expected Outputs

Teams should present:

a) RUL Prediction Algorithm / Model
b) Graphical representation of battery degradation
c) Estimated remaining cycles or months of operation
d) Sensitivity analysis showing impact of:

    i. temperature
    ii. depth of discharge
    iii. charging rate

e) Visualization dashboard or simulation (optional)

8. Evaluation Criteria

Solutions will be evaluated based on the following parameters:

Criteria	Weightage
Technical approach and innovation	25%
Prediction accuracy and robustness	25%
Practical deployability in EV/BMS systems	20%
Quality of analysis and visualization	15%
Business relevance and scalability	15%

MIDC Speed vs. time profile (Annexure – A)

Total Cycle Duration

- **Cycle time:** ~1080 seconds (18 minutes)
- **Average speed:** ~27 km/h
- **Max speed:** ~60 km/h
- **Driving nature:** Urban stop-and-go

Time (s)	Vehicle Speed (km/h)
0	0
10	10
20	20
30	30
40	35
50	30
60	25
70	0
90	15
110	30
130	40
150	50
170	45
200	35
230	25
260	10
300	0
340	20
380	35
420	45
460	55
500	60
540	55
580	45
620	35
660	20
700	0
760	20
820	35
880	50
940	60
1000	40
1060	20
1080	0